Tree Host

Minecraft server hosting · treehost.online

INFRASTRUCTURE · MIGRATION · LAW — INTERNAL PLAN, OCTOBER 2026

Going live: from launch fund to real customers

Which bare-metal nodes and VPSes Tree Host should rent, what each one earns at our prices, the step-by-step migration that puts treehost.online on a production server and opens it to paying customers, and the laws that apply to us as a Minecraft hosting company.

OVH RISE-S	25	17	~4 weeks
Recommended first production node (Ryzen 7 9700X, 64 GB)	Sapling servers per 64 GB node, after overhead	Paying Saplings to break even on RISE-S	From ordering the server to public launch

Prepared for the Tree Host team

Prices checked 2 October 2026. Not legal advice.

AT A GLANCE

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1 Executive summary

Tree Host is built and works end to end: website, dashboard, Stripe billing, Firebase accounts and one Docker container per customer server. The remaining work is renting the right machine, moving the site onto it, and making the business legally ready to take money.

The recommendation

- **Production node:** OVHcloud **RISE-S** in Vint Hill, Virginia: AMD Ryzen 7 9700X (8 cores up to 5.5 GHz), 64 GB DDR5 ECC, 2 × 512 GB NVMe, OVH's always-on anti-DDoS. **$77/month + $77 setup.**
- **If the fund can't cover $154 up front:** start on **netcup RS 4000 G12** (32 GB, 12 dedicated cores, about $33/month, no setup fee) and move to RISE-S once monthly revenue passes about $80.
- **Off-site backups:** a storage box or S3 bucket at another company (a few dollars a month).
- **Later:** a second node per region once multi-node is built (roadmap), with a premium Ryzen X3D tier priced higher.

What changed in 2026 (and why it matters)

- DDR5 RAM costs roughly **6×** what it did a year ago (OVHcloud). Hetzner raised prices on **15 June 2026**; OVH raised its Advance, Game and Scale ranges on **11 August**, with renewals following on **1 October**.
- OVH's budget **Eco ranges (Kimsufi, So you Start, Rise) were not raised**, which makes Rise the best value per GB for us.
- At our price of **$2.50 per GB** (Sapling), premium Ryzen X3D machines now **lose money**. We need nodes at about **$1.20 per GB** or less.
- Oracle's Always Free ARM allowance dropped from 24 GB to **12 GB**. Our hosting guide still says 24 GB.

The eight things to do first

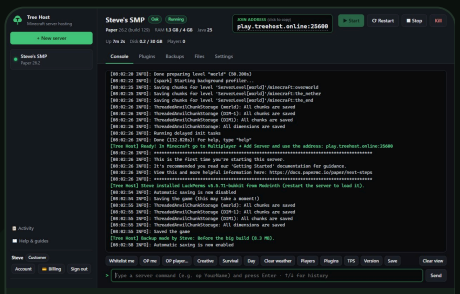
#	ACTION	WHY	OWNER
1	**Form a legal entity** (an LLC or the local equivalent) and open a business bank account	Stripe live mode, provider contracts and liability protection all need a real business owned by an adult (18+).	Founder

#	ACTION	WHY	OWNER
2	**Fix the capacity gap for multi-server plans** (see §6)	Monthly plans allow unlimited servers, but capacity only reserves memory for one. A single customer could fill a node.	Backend
3	**Order RISE-S** (or netcup RS 4000 if cash is short) in a US-East data centre	Best value per GB in 2026 with real anti-DDoS, close to North American players.	DevOps
4	Run `install.sh`, point DNS, get `doctor.sh` all green	The repo already automates Docker, Node, Caddy, firewall and backups.	DevOps
5	Set **Stripe live keys**, the webhook, **Stripe Tax** and the Customer Portal	Taking real payments, collecting tax correctly, and letting customers cancel online.	Admin
6	Fill in `TREEHOST_JURISDICTION` and update the Terms and Privacy policy	The Terms currently say "governed by the laws of *your jurisdiction*", a template placeholder.	Founder
7	Register a **DMCA agent** ($6), add an **age gate** and a **refund/withdrawal** clause	Copyright safe harbour, COPPA, EU/UK consumer law.	Founder
8	Soft launch to Discord members, then go public	Find problems with 5–10 friendly customers before strangers find them.	Everyone

2 Where Tree Host is today

To pick hardware we first need to know what the software asks of it. Everything below comes from the current repository (`src/`, `deploy/`, `staff-docs/hosting-guide.md`).

```
Players ── play.treehost.online:25600-25999 ──┐
                                              ▼
Customers ─ https://treehost.online ─► Production node (Ubuntu 24.04)
                                       ├─ Caddy: HTTPS, auto certificates
                                       ├─ Tree Host (Node.js 22): site,
                                       │   dashboard, API, billing, waker
                                       ├─ Docker: 1 container / server
                                       │   (memory, CPU, PID limits,
                                       │    read-only, no capabilities)
                                       └─ nightly backup → off-site rsync
Stripe ── payments + webhook ──────────┘
Firebase ── sign-in + Realtime Database (accounts, plans)
Resend ── payment and notice emails
```



The customer dashboard: live console, join address, plan, RAM and disk.

- **One machine runs everything** today. Multi-node is on the roadmap but not built, so the first node must be big enough to grow into.
- Each server gets its own **port** from `TREEHOST_PORTS=25600-25999` (400 ports per node).
- Idle servers can **sleep** (`src/waker.js`): they free their RAM and wake when a player joins.

What we sell (default plans in `src/billing.js`)

PLAN	RAM	CPU	DISK	PRICE	RESERVED ON NODE*	PRICE PER SOLD GB	AIMED AT
Sapling	2 GB	1	15 GB	$5 / mo	2.38 GB	$2.50	1–5 players, small survival

PLAN	RAM	CPU	DISK	PRICE	RESERVED ON NODE*	PRICE PER SOLD GB	AIMED AT
Oak	4 GB	2	30 GB	$9 / mo	4.38 GB	$2.25	5–20 players, light modpacks
Redwood	8 GB	3	60 GB	$16 / mo	8.38 GB	$2.00	20–60 players, big modpacks
Lifetime	4 × 4 GB + 2 × 1 GB	2	30 GB	$500 once	20.25 GB	—	4 servers, 2 proxies, priority support

* Reserved = plan RAM + 384 MB Java overhead per server (`OVERHEAD_MB`). The node's sellable capacity defaults to total RAM − 2 GB and can be set in the Staff Console.

So the node must provide

💾 Memory first
RAM is literally what we sell. Every other resource is shared; memory is reserved. ECC is strongly preferred: a memory error in a customer's world is a support ticket we can't fix.

⚡ Fast single cores
A Minecraft server's main tick loop runs on one thread. A 5+ GHz Ryzen holds 20 TPS with far more players than a 3 GHz server CPU with many slow cores.

🛡️ DDoS that understands games
Public Minecraft servers get attacked constantly. Filtering must cover the whole 25600–25999 range, not just the website.

🗄️ NVMe storage
Chunk loading, world saves and nightly backups are disk-heavy. Two drives in RAID 1 survive a drive failure.

🌐 Bandwidth & IPs
About 1 Gbps unmetered. One IPv4 address is enough because servers are told apart by port.

📜 Terms that allow us
The provider must allow game servers and hosting for third parties (reselling). All providers below do; their abuse desks still expect replies within 24–72 hours.

3 The 2026 hardware market: the RAM price shock

Demand for high-bandwidth memory in AI accelerators has squeezed DDR5 and NVMe supply. OVHcloud reported that DDR5 now costs about six times what it did a year earlier. Since memory is what we sell, this decides which providers work for us.

PROVIDER	WHEN	WHAT CHANGED	EFFECT ON US
Hetzner	15 June 2026 (the third rise of 2026)	New model names (AX42-1, AX102-1…), monthly prices up, setup fees down. AX42-1: **€97.30** ($117.10) + €49 setup. AX102-1: **€257.30** ($302.10). Cloud CCX13 went from €15.99 to **€42.99**.	**Too expensive per GB** for our current prices. The auction is still worth watching.
OVHcloud	1 Jul (options), 11 Aug (new orders), 1 Oct 2026 (renewals)	Advance, Game, Scale and High Grade ranges up 16–74%, and RAM upgrades up 68–652%. **Kimsufi, So you Start and Rise unchanged.**	**Rise is the sweet spot** Order the base RAM configuration; avoid paid RAM upgrades.

PROVIDER	WHEN	WHAT CHANGED	EFFECT ON US
Oracle Cloud	Mid-2026	Always Free ARM allowance cut from 4 OCPU / 24 GB to **2 OCPU / 12 GB** (reports say Pay-As-You-Go accounts may keep 4/24).	**Staging only** Update `staff-docs/hosting-guide.md`.
netcup	—	Root Server G12 line (AMD EPYC, DDR5 ECC, dedicated cores) is still priced low per GB.	**Best budget start**

What this means

Lock in hardware now on a range that wasn't repriced, take the base RAM configuration, and treat any annual prepay discount as a hedge against more increases. Don't build the business on a node whose renewal price can jump. OVH renewals went up on 1 October 2026 for the affected ranges.

4 What makes a good Tree Host node

Each provider in §5 is scored against these criteria. The weights reflect what customers notice (lag, downtime) and what keeps us profitable (cost per GB).

CRITERION	WEIGHT	GOOD LOOKS LIKE	RED FLAG
Cost per GB of RAM	25%	≤ $1.20 per GB per month including setup spread over 12 months	> $1.80 per GB at our $2–2.50 price points
Single-thread speed	20%	Ryzen 7000/9000 desktop-class, boost ≥ 5 GHz; X3D cache is a bonus	Shared vCPUs, old Xeons, < 3.5 GHz boost
Game-aware DDoS	20%	Always-on filtering that covers UDP/TCP game ports, no extra fee	"Null-route on attack": your whole node goes offline
Location / latency	10%	US-East (Virginia) for North America; Germany/France for Europe	One location only, far from our players
Reliability	10%	ECC RAM, RAID 1 NVMe, IPMI/KVM console, SLA ≥ 99.9%	No out-of-band console, no hardware SLA
Contract & terms	10%	Monthly term, reselling allowed, clear abuse process	12-month lock-in, "no game servers" or "no resale"
Upfront cost	5%	No or low setup fee (we have a $50 launch fund)	Setup fee ≥ one month of rent

5 Provider shortlist: bare metal and VPS

Prices are list prices seen in early October 2026, before tax. Euro prices are shown as listed; for budgeting we use about $1.17 per €1. Stock and prices change often, so check before ordering.

5.1 Dedicated bare-metal servers

SERVER	CPU	RAM	DISK	MONTHLY	SETUP	$/GB	DDOS	FIT
OVH RISE-S ★ Eco · US, CA, EU	Ryzen 7 9700X 8c/16t, 3.8–5.5 GHz	64 GB DDR5 ECC	2×512 GB NVMe	$77	$77	$1.20	OVH Anti-DDoS (always on)	Best value
OVH RISE-Game-1 Eco · game range	Ryzen 5 5600X 6c/12t, 3.7–4.6 GHz	32–64 GB DDR4 ECC	2×512 GB NVMe	from $90	$90	$2.81*	**Game DDoS** protection	Strong DDoS, weaker CPU
OVH RISE-1 / RISE-2	Xeon-E 2386G / 2388G	32–128 GB DDR4 ECC	2×512 GB+ NVMe	$70 / $80	= 1 mo	$2.19*	OVH Anti-DDoS	OK, older CPU
OVH Game-1 (Gen 2026)	Ryzen 7 9800X3D	64–256 GB DDR5 ECC	2×960 GB NVMe	from $357	$357	$5.58	Game DDoS	Premium tier only
OVH Game-2 (Gen 2026)	Ryzen 9 9950X3D	64–256 GB DDR5 ECC	2×960 GB NVMe	from $357	$401	$5.58	Game DDoS	Premium tier only
Hetzner AX42-1 Germany / Finland	Ryzen 7 PRO 8700GE	64 GB DDR5 ECC	2×NVMe	€97.30	€49	$1.83	Basic network filter	EU node later
Hetzner AX102-1	Ryzen 9 7950X3D	128 GB DDR5 ECC	2×1.92 TB NVMe	€257.30	€129	$2.36	Basic network filter	Too costly now
Hetzner Server Auction	Ryzen 5 3600-class (older)	64 GB DDR4	2×512 GB NVMe	~€40–55	€0	~$0.85	Basic network filter	Cheap EU node
US Ryzen 9 specialists (e.g. Tier.Net)	Ryzen 9 7950X / 9950X	64–128 GB	NVMe	$289–329	varies	$2.5–5	Game filtering included	Premium tier only

* Base configuration has 32 GB; $/GB is for the base price. RAM upgrades add cost, and on the repriced OVH ranges RAM options rose sharply.

5.2 VPS and "root servers" (dedicated cores, virtualised)

SERVER	VCPU	RAM	DISK	MONTHLY	$/GB	NOTES	FIT
netcup RS 4000 G12 ★	12 dedicated EPYC	32 GB DDR5 ECC	1 TB NVMe	€33.55 incl. 19% VAT (≈€28.19 net)	$1.03	No setup fee; EU and US (Manassas, VA) locations	Launch on $50
netcup RS 8000 G12	16 dedicated EPYC	64 GB DDR5 ECC	2 TB NVMe	€59.97 incl. VAT (≈€50.39)	$0.92	Cheapest 64 GB with guaranteed cores	Great value
netcup RS 2000 G12	8 dedicated	16 GB	512 GB NVMe	€18.01 incl. VAT	$1.11	Good staging / web front-end box	Staging
Contabo Cloud VDS M	8 dedicated	32 GB	240 GB	€49	$1.79	Small disk for 15–60 GB plans	Meh
Contabo Cloud VDS XL	16 dedicated	64 GB	480 GB	€94	$1.72	Disk too small for 25 servers	Meh
Hetzner Cloud CCX13/23	2 / 4 dedicated	8 / 16 GB	80 / 160 GB	€42.99 / €85.99	$6.29	Repriced up to 2.7× in June 2026	Avoid
Oracle Always Free (ARM)	2 OCPU (Ampere)	12 GB	200 GB	$0	$0	Capacity is scarce; idle instances can be reclaimed	Staging only

⚠ **The catch with VPS and EPYC**

netcup's G12 uses EPYC cores with a lower boost clock than a desktop Ryzen, so single-thread speed is roughly a third lower. That's fine for 1–20 player Sapling and Oak servers, but heavy modpacks feel it. netcup's DDoS filtering is basic, which is the main reason we treat it as a **launch step**, not the final home.

5.3 Provider profiles

OVHcloud (France · US: Vint Hill VA, Hillsboro OR · Canada: Beauharnois)

The default choice for game hosts. Always-on anti-DDoS (VAC) on every server, and the stronger **Game DDoS** filter on the game ranges. The Eco/Rise range was **not** repriced in 2026. US servers are sold by OVH US LLC, the separate US company.

+ DDoS, price per GB, locations, IPMI. – Setup fee equals one month, support is slow on budget ranges, and Game-range prices rose.

netcup (Germany/Austria · Manassas, Virginia)

Very low price per GB with dedicated cores and DDR5 ECC. A good place to launch on a $50 budget, or later as a web/staging box.

+ Price, no setup fee, big NVMe. – Basic DDoS, slower cores, some plans have minimum terms (check before ordering).

Hetzner (Germany · Finland)

Excellent hardware and network quality in the EU, but after three rises in 2026 new dedicated servers cost too much per GB for our plans. The **Server Auction** (older Ryzen, 64 GB, no setup fee) is the exception and makes a good future EU node. DDoS filtering is basic and can be slow to react to game attacks.

Oracle Cloud (Always Free)

Free ARM capacity that's now smaller (12 GB). Use it as a **staging server** that mirrors production, for testing updates before running `update.sh` on the live node. Don't run paying customers on it: free capacity can be reclaimed and isn't backed by an SLA.

6 Capacity & unit economics

How many servers each node holds and what it earns. Assumptions: 4 GB of each node is kept for Ubuntu, Docker, Node.js, Caddy and file cache; each server reserves its RAM plus 384 MB; Stripe takes 2.9% + $0.30 per US card payment; "75% full" is the realistic steady state.

6.1 Servers per node

NODE RAM (USABLE)	SAPLING (2.38 GB)	OAK (4.38 GB)	REDWOOD (8.38 GB)	LIFETIME BUNDLES (20.25 GB)
12 GB Oracle (8 GB)	3	1	0	0
32 GB (28 GB)	11	6	3	1
64 GB (60 GB)	25	13	7	2
128 GB (124 GB)	52	28	14	6

6.2 What each node earns (all Sapling, after Stripe fees of $0.45 each)

NODE	COST / MO	SLOTS	NET REVENUE FULL	NET AT 75%	PROFIT AT 75%	BREAK-EVEN	VERDICT
netcup RS 4000 G12 (32 GB)	≈ $33	11	$50.11	$36.44	**+$3**	8 customers	Launch
netcup RS 8000 G12 (64 GB)	≈ $59	25	$113.88	$81.99	**+$23**	13 customers	Budget scale

NODE	COST / MO	SLOTS	NET REVENUE FULL	NET AT 75%	PROFIT AT 75%	BREAK-EVEN	VERDICT
OVH RISE-S (64 GB)	$77	25	$113.88	$81.99	**+$5**	17 customers	Production
Hetzner auction (64 GB)	≈ $55	25	$113.88	$81.99	**+$27**	13 customers	EU node
Hetzner AX42-1 (64 GB)	$117	25	$113.88	$81.99	**−$35**	never (26 > 25)	No
Hetzner AX102-1 (128 GB)	$302	52	$236.86	$177.65	**−$125**	never	No
OVH Game-2 (64 GB)	$357	25	$113.88	$81.99	**−$275**	never	No

Per reserved GB, Oak earns about the same as Sapling ($1.93 vs $1.92 net) and Redwood a little less ($1.82), so a mix of plans gives similar results. Setup fees are not included above; RISE-S's $77 setup costs about $6.40 a month if spread over a year.

🌱 **Pricing insight: add a premium tier later instead of raising prices**

Our $2.00–2.50 per GB is in line with the budget end of the market (rivals advertise from $1/GB up to ~$3.50/GB for Ryzen 9950X). Keep the current plans on Rise-class nodes. When we add X3D hardware, sell it as a separate **"Performance"** line at about **$4.50 per GB** (e.g. 8 GB for $36), on 128 GB hardware that costs $3 per GB or less. Example: a Hetzner AX102-1 (7950X3D, 128 GB, $302) holds 14 of those servers, about **$485 net** when full. A 64 GB OVH Game-2 at $357 fits only 7, so it doesn't pay even at premium prices.

6.3 Two problems in the current code that cost money

🛑 **Monthly plans allow unlimited servers, but only one is counted toward capacity**

`slotError()` in `src/billing.js` lets a monthly plan run any number of servers, each with the plan's full RAM, while `soldMb()` reserves memory for **one** server per subscription. Ten customers each running five 4 GB Oak servers would ask for 200 GB on a 64 GB node: the Linux OOM killer would start stopping customers' servers. **Fix before launch:** (a) count *running* servers' memory against capacity and refuse to start when the node is full, (b) turn sleep on by default for extra servers, or (c) limit a monthly plan to one running server (others can sleep). Option (a) is the safest.

⚠️ **The Lifetime plan permanently uses a third of a 64 GB node**

$500 once holds 20.25 GB forever. On RISE-S that's about $24 of rent every month, so it breaks even after about **20 months** and loses money after that, with no end date. Recommend: limit sales (e.g. 2 per node, which the "sold out" capacity check already allows), raise the price to at least $750, or make it "lifetime of the product with a 5-year minimum" in the Terms.

7 DDoS protection & networking

Minecraft is one of the most-attacked games online. Kids with stresser subscriptions hit servers after being banned. Every Tree Host customer shares one IP, so an attack on one server reaches all of them.

Layer 1: provider network
OVH's VAC scrubs volumetric floods for free, always on. The Game DDoS filter on the game ranges adds per-protocol rules. This is the main reason to prefer OVH for production.

Layer 2: Minecraft proxy (optional)
TCPShield (free 1 TB, Pro $25/mo) or NeoProtect (free tier, Universal €15/mo) filter bots and L7 floods. They route by **hostname**, so Tree Host would need per-server subdomains instead of ports.

Layer 3: on the node
ufw for the host, Docker limits per server, and connection throttles in `server.properties` and Velocity. Never expose the panel's port 8766: Caddy already proxies it.

Recommended network setup

- **Keep the website separate from game traffic:** put `treehost.online` and `www` behind Cloudflare's proxy (orange cloud, set `TREEHOST_TRUST_CLOUDFLARE=1`) so the panel's real IP stays hidden. `play` must stay **DNS only** (grey cloud).
- **Order a second IPv4 address** (failover/additional IP) for the panel when the node supports it, so a flood at `play` doesn't reach the dashboard.
- **OVH firewall:** in the OVH control panel, enable the Network Firewall / Game firewall with rules for TCP 22 (your IP only), 80, 443 and 25600–25999. Mirror the port range exactly with `TREEHOST_PORTS`.
- **Later: SRV records.** Creating `_minecraft._tcp.<name>.play.treehost.online` per server lets players join `steve.play.treehost.online` without the port, the same thing big hosts do.

⚠ **Docker and ufw gotcha**
Ports that Docker publishes skip ufw's rules. That's fine for 25600–25999, which should be open anyway, but never publish any other container port (databases, RCON) on 0.0.0.0. Tree Host's containers only publish the game port; keep it that way when adding features.

8 Recommended growth path

Spend money only when revenue justifies it. Each phase says what triggers the next one.

Phase 0
Now · $0/mo

Staging on Oracle Always Free (12 GB) or a local PC.
- Install exactly like production with `install.sh`, using Stripe **test** keys on a `staging.treehost.online` subdomain.
- Rehearse the whole migration runbook (§9) here first, including restoring a backup.
- **Next when:** the entity exists and Stripe is approved for live payments.

Phase 1
Launch · $33–77/mo

One production node: OVH RISE-S (preferred) or netcup RS 4000 G12 (if cash is under $154).
- Up to 25 Saplings (RISE-S) or 11 (netcup). Set the Staff Console capacity to about 90% of RAM.
- Nightly off-site backups. Free uptime monitoring. Soft launch to Discord first.
- **Next when:** capacity is above 70% for two weeks, or monthly revenue is above $80 (on netcup).

Phase 2
Growing · $90–170/mo

Build multi-node (roadmap), split the web panel onto its own small VPS, then add a second game node.

- Web/API on a netcup RS 2000 (about €18). Game nodes run a small Tree Host agent that only manages Docker.
- Second node: another RISE-S in the same region, or a Hetzner auction 64 GB box in Germany for EU players.
- **Next when:** there are 40+ paying customers, or customers ask for heavy modpacks or 60+ player servers.

Phase 3
Scale · $400+/mo

Add a premium "Performance" line on 128 GB Ryzen X3D hardware (e.g. Hetzner AX102-1 for the EU, or a US Ryzen 9 specialist), priced at about $4.50 per GB.

- Location choice at checkout (US-East, EU). Status page. 24/7 rota for support.
- Consider a professional review of contracts and policies, plus business insurance (cyber/E&O).

9 Migration runbook: from test site to live platform

The test site (`Start.bat`, `test-site-data/`, Stripe test keys) becomes a real production deployment. **Don't copy test data**: start production with a clean `/var/lib/tree-host`. Accounts and plans live in Firebase, so use a fresh production database or clear the test records.

Week 1: Business and accounts (no servers yet)

#	TASK	DETAILS
1	Legal entity & bank	Form an LLC (or local equivalent), get a tax ID (EIN in the US), open a business bank account. The owner must be 18+.
2	Stripe account activation	Business details, bank account, statement descriptor **TREEHOST.ONLINE**, support email/phone, website URL. Turn on **Stripe Tax** and **Radar**. In *Customer portal*, allow cancellations and card updates.
3	Email	Inboxes for `support@`, `billing@` and `promotion@` (Cloudflare Email Routing is free and forwards to a shared mailbox). Resend account: verify the domain (SPF, DKIM records), then add a DMARC record.
4	Firebase production	Authentication → *Authorized domains*: add `treehost.online`. Customise the email templates (sender name "Tree Host", link domain). Deploy `deploy/database.rules.json`. Generate the **service account key** and keep it off every PC once uploaded.
5	Order the node	OVH RISE-S → Vint Hill (US-East), Ubuntu 24.04, your SSH public key. Note the IPv4/IPv6 addresses and the IPMI login.
6	Backup target	Storage at a **different company** from the node (e.g. a Hetzner Storage Box or Backblaze B2) for `TREEHOST_BACKUP_RSYNC`.

Week 2: Build the node

```
# 1. First login: update, create an admin user, lock down SSH
ssh root@NODE_IP
apt update && apt -y full-upgrade && reboot
adduser ops && usermod -aG sudo ops && rsync -a ~/.ssh /home/ops/ && chown -R ops: /home/ops/.ssh
sed -i 's/^#\?PasswordAuthentication.*/PasswordAuthentication no/; s/^#\?PermitRootLogin.*/PermitRootLogin
prohibit-password/' /etc/ssh/sshd_config
systemctl restart ssh && apt -y install fail2ban

# 2. Get the code (private repo, read-only deploy key) and install
sudo git clone https://github.com/ItsJeffTheDev/Honey-Host.git /opt/tree-host
sudo bash /opt/tree-host/deploy/deploy-key.sh       # one time: future pulls without a token
sudo bash /opt/tree-host/deploy/install.sh          # Docker, Node 22, Caddy, ufw, systemd, backups

# 3. Secrets: Firebase key + settings (root:treehost, mode 640)
scp tree-host-firebase.json ops@NODE_IP:/tmp/ && sudo mv /tmp/tree-host-firebase.json /etc/
sudo nano /etc/tree-host.env       # fill in the values from the table below
sudo systemctl restart tree-host && sudo bash /opt/tree-host/deploy/doctor.sh
```

SETTING IN `/ETC/TREE-HOST.ENV`	PRODUCTION VALUE
`TREEHOST_PUBLIC_URL` / `TREEHOST_GAME_HOST`	`https://treehost.online` / `play.treehost.online`
`TREEHOST_PORTS`	`25600-25999`, the same range as the OVH firewall rules
`TREEHOST_ADMIN_EMAILS`	The owner's email (becomes Owner once verified)
`TREEHOST_COMPANY` / `TREEHOST_JURISDICTION`	The **legal entity name** (e.g. "Tree Host LLC") / the state and country it's registered in. **Currently empty.**
`STRIPE_SECRET_KEY` / `STRIPE_WEBHOOK_SECRET`	`sk_live_…` / the secret from the webhook endpoint `https://treehost.online/api/billing/webhook`
`BILLING_DEMO`	**0** (never 1 on a live site)
`RESEND_API_KEY` / `MAIL_FROM`	The live key / `Tree Host <no-reply@treehost.online>`
`TREEHOST_BACKUP_RSYNC` / `…_KEEP`	The off-site target / 7 (a week of nightly copies)
`TREEHOST_TRUST_CLOUDFLARE`	1 only if the website records use Cloudflare's orange cloud

Week 2 (continued): DNS for treehost.online

TYPE	NAME	VALUE	PROXY	PURPOSE
A / AAAA	`@`	Node IPv4 / IPv6	Orange (optional)	Website and dashboard
A / AAAA	`www`	Node IPv4 / IPv6	Orange (optional)	Redirects to treehost.online (Caddy)
A / AAAA	`play`	Node IPv4 / IPv6	**DNS only**	Players join play.treehost.online:PORT
A	`staging`	Oracle/staging IP	DNS only	Test new versions before production

TYPE	NAME	VALUE	PROXY	PURPOSE
TXT / CNAME	(from Resend)	SPF, DKIM	—	Payment emails land in inboxes, not spam
TXT	`_dmarc`	`v=DMARC1; p=quarantine; rua=mailto:support@treehost.online`	—	Stops others spoofing our domain
MX	`@`	Your mail or forwarding provider	—	Receiving support@, billing@
CAA	`@`	`0 issue "letsencrypt.org"`	—	Only Caddy's certificate authority can issue certificates

Week 3: Test like a customer

- [ ] **doctor.sh** all green: services, DNS, HTTPS, firewall, disk, backups.
- [ ] Sign up with a new email: verification email arrives, Google sign-in works on `treehost.online`.
- [ ] Buy each plan with a **real card**, then refund it from Stripe. Check the webhook marks plans active and refunded ones cancelled.
- [ ] Create one server of each type (Paper, Fabric, Velocity…). Join from a real Minecraft client over the internet.
- [ ] Fill the node: start servers until "sold out" appears. Confirm no server is OOM-killed (fix from §6.3 in place).
- [ ] Failed payment: in Stripe test mode with a test clock, check the 3-day grace period, the suspension and the email.
- [ ] Run `systemctl start tree-host-backup` and **restore it** on the staging box. A backup you haven't restored isn't proven.
- [ ] Reboot the node: everything (Caddy, Tree Host, auto-start servers) comes back by itself.
- [ ] Moderation: suspend and unsuspend a test account; the appeal flow and email work.
- [ ] Legal pages updated (see §12) and linked in the footer and checkout.

Week 4: Soft launch → public launch

Soft launch (days 1–7)

- Open sign-ups to Discord members only, with a launch promo code.
- Staff watch `journalctl -u tree-host -f` and the Discord #status channel.
- Fix every rough edge before more people arrive.

Public launch (day 8+)

- Announce on YouTube/Discord; tick "First paying customer 🎉" on the roadmap.
- Keep capacity at 90% so "sold out" shows before the node is overloaded.
- Order the next node when capacity is above 70% for two weeks.

🌱 **Moving to a bigger node later (netcup → OVH, or any upgrade)**
Install the new node with `install.sh`, stop Tree Host on the old one, `rsync -aHAX /var/lib/tree-host/` and `/etc/tree-host*` across, start the new node, then switch the DNS A records (set TTL to 300 s a day before). Player addresses stay the same because servers keep their ports. Plan for 15–30 minutes of downtime and announce it 48 hours ahead.

10 Operations after launch

Paying customers expect their server to be up when their friends log on.

📈 Monitoring

- Free external checks (e.g. UptimeRobot or Better Stack) on `https://treehost.online` and one canary Minecraft server port, alerting Discord.
- Daily look at the Staff Console: sold capacity, disk, failed payments.
- Alert when disk is above 80%: worlds grow, backups grow faster.

💾 Backups (3-2-1)

- Customer backups on the node (dashboard), nightly full archive, and an **off-site copy** at a different company.
- Keep 7 nightly copies; test one restore every month.
- Also back up `/etc/tree-host.env` and the Firebase key to a password manager.

🔄 Updates

- Ship to staging first, then `update.sh` on production in a quiet hour (servers stop cleanly).
- `unattended-upgrades` covers security patches. Reboot monthly in a maintenance window that has been announced in advance.

🎧 Support & abuse

- Targets: first reply in 12 h, priority (Lifetime) in 4 h.
- **Answer the provider's abuse emails within 24 h**. OVH and netcup can block an IP if reports are ignored.
- Use the moderation tools (action numbers, appeals) for every enforcement action, and keep the evidence.

Monthly business check

METRIC	TARGET	ACTION IF MISSED
Node cost ÷ net revenue	< 60%	Run promotions, review prices, check that free plans and giveaways aren't eating capacity
Sold capacity	60–85%	Under: marketing push. Over: order the next node
Uptime (website and canary port)	≥ 99.5%	Write a post-incident note in Discord; credit affected customers
Chargebacks / disputes	< 0.5% of payments	Stripe can close accounts above about 1%. Improve the cancel flow and refunds

11 Laws that apply to us as a Minecraft hosting company

We bill in US dollars and sell worldwide online, so this section assumes a **US-based business with customers in the US, EU, UK and elsewhere**. Where you register the business changes the details. **This is a planning guide, not legal advice**: have a lawyer in your state or country review the Terms and Privacy policy before launch (the README already says so).

11.1 Mojang / Microsoft: the Minecraft EULA and Usage Guidelines

⚖️ **Applies from day one, to us and to every customer**

- **EULA acceptance must be the customer's.** We may not accept it for them. ✓ Tree Host already requires the customer to tick "I agree to the Minecraft EULA" before a server starts (`src/instances.js`, `src/server.js`), and writes a note that it was accepted through Tree Host.
- **Branding:** we may say "Minecraft server hosting" to describe the service, but we must not use the Minecraft logo, imply that Mojang endorses us, or put "Minecraft" in our company or product name. ✓ The footer already says "*Not an official Minecraft product. Not approved by or associated with Mojang or Microsoft.*" Keep it on every page and in ads.
- **Selling hosting is allowed**: we sell server infrastructure, not the game. We must not sell Minecraft accounts, give access to the game itself, or resell Mojang content (skins, capes, official assets).
- **Customers' own monetisation** (ranks, cosmetics) is governed by the Usage Guidelines. In August 2026 Mojang merged the Commercial Usage Guidelines into the Usage Guidelines as a "commercial use" section with a servers subsection. Rules include: no pay-to-win advantages sold to players, no gambling or paid loot boxes with gameplay value, no NFTs/blockchain, and a disclaimer on server stores. Our Terms already forbid breaking the EULA and Usage Guidelines; when a customer's server gets reported, we act through our moderation flow.
- Mojang can change or withdraw the guidelines at any time. Assign someone to check `minecraft.net/usage-guidelines` every quarter.

11.2 Consumer protection, subscriptions and refunds

LAW	WHERE	WHAT IT REQUIRES OF US
FTC Act §5 (unfair/deceptive practices)	US, federal	Advertising must be true: "DDoS protection", "99.9% uptime", "unlimited", "lifetime" must match reality. The **Lifetime** plan must say clearly what happens if the service ends.
ROSCA (Restore Online Shoppers' Confidence Act)	US, federal	Before charging for a renewing plan: show all material terms clearly, get **express informed consent**, and give a **simple way to cancel**. The FTC's 2024 "click-to-cancel" rule was vacated by the 8th Circuit on 8 July 2025, but the FTC has started a new rulemaking and ROSCA still applies.
California Automatic Renewal Law (as amended by AB 2863, in force 1 July 2025)	CA customers (in practice: everyone)	Clear renewal terms next to the "subscribe" button; an **affirmative consent** step; an acknowledgement email with how to cancel; **online cancellation** through the same channel; notice before price changes; keep proof of consent for **3 years**. Many other states (NY, VA, CO, IL…) have similar laws.
EU Consumer Rights Directive / UK Consumer Contracts Regulations	EU / UK customers	**14-day right of withdrawal** for online services. To start the server immediately, the customer must **expressly ask for immediate start and acknowledge** they lose the full right (pro-rata refund if they withdraw). Our Terms' "generally non-refundable" needs an EU/UK carve-out.
Australian Consumer Law	AU customers	"No refunds" statements are illegal; consumer guarantees (service fit for purpose) can't be excluded.

11.3 Privacy and children

LAW	APPLIES WHEN	WHAT IT MEANS FOR TREE HOST
COPPA (Children's Online Privacy Protection Act), amended rule in force 23 June 2025, **full compliance from 22 April 2026**	We have actual knowledge of users under 13, or the service is directed at children	Minecraft's audience skews young, so this is our biggest privacy risk. Options: (1) **age gate at sign-up** and refuse under-13 accounts (recommended), or (2) verifiable parental consent, a written data-retention policy and a security programme. Paid accounts should require 18+ or a parent as the account holder. The current privacy line ("a parent or guardian must agree") isn't enough on its own for under-13s.
GDPR (EU) and **UK GDPR**	From the first EU/UK customer (Art. 3(2): we offer services to them)	Lawful basis per purpose, privacy notice (ours is a good start), data-subject rights within 1 month, **Data Processing Agreements** with Firebase/Google, Stripe, Resend and the host, international transfer safeguards (EU-US Data Privacy Framework / SCCs), breach notification to the regulator within **72 h**. A non-EU business usually also needs an **EU and UK representative** (Art. 27). Age of digital consent is 13–16 depending on the country.
We're also a processor for our customers	Always	Customers' server logs contain their *players*' names and IPs. Offer a short **DPA** to customers and describe this in the Terms.
CCPA/CPRA and ~20 other US state privacy laws	Mostly above 100,000 residents' data or $25M revenue (Texas: any non-small business; Connecticut's threshold drops to 35,000 in July 2026)	Probably not triggered at launch, but the same practices (notice, access/delete, no selling data) cover them. Keep "we don't sell your data" true.
Data breach notification laws	All 50 US states, plus GDPR	An incident plan: who decides, who notifies customers and regulators, and within what time limit. Keep access logs.

11.4 Hosting other people's content

LAW	WHERE	WHAT WE MUST DO
DMCA §512 safe harbour	US	Customers upload plugins, mods and worlds that may infringe copyright. To be protected: **register a designated agent** with the US Copyright Office ($6, **renew every 3 years**), publish the agent's contact details on the site, act on valid takedown notices, handle counter-notices, and have a **repeat-infringer termination policy** in the Terms.
Section 230 (Communications Decency Act)	US	Generally shields us from liability for content our customers create. It doesn't cover federal crimes or intellectual property (that's what the DMCA is for).
18 U.S.C. §2258A (CSAM reporting)	US	As a remote computing service, if we gain actual knowledge of child sexual abuse material we **must report it to NCMEC's CyberTipline** and preserve the evidence. Never copy or forward it. Write this into the support playbook.
Stored Communications Act	US	Rules for law-enforcement requests: subpoena vs. court order vs. warrant. One named person handles them, and requests are logged.

LAW	WHERE	WHAT WE MUST DO
EU Digital Services Act (hosting service rules)	EU users	All hosting providers: a **single point of contact** for authorities and users, clear content rules in the Terms, a **notice-and-action** mechanism, a **statement of reasons** for every restriction, and reporting threats to life. Non-EU providers need an **EU legal representative**. ✓ Our moderation already sends reasons with action numbers and allows appeals. Micro/small businesses are exempt from many of the heavier online-platform duties, but not from these.
UK Online Safety Act	UK users	Customers' servers with public chat are "user-to-user" services; Tree Host is mainly their infrastructure. Watch item: get advice before targeting the UK heavily.

11.5 Payments, tax and money

AREA	WHERE	WHAT APPLIES
PCI DSS	Card payments	Stripe Checkout means card data never touches our server, so we only need the short **SAQ A** self-assessment. Never log or store card numbers.
US sales tax	Per state	Since *Wayfair* (2018), states can tax remote sellers once sales pass "economic nexus" thresholds (often $100k). Hosting/SaaS is taxable in some states and exempt in others. Stripe Tax monitors thresholds and collects where needed. Register in your home state first.
EU VAT (non-Union OSS)	EU consumers	For B2C digital services from a non-EU seller there's **no threshold**: VAT is due at the customer's country rate from the first sale. Register for non-Union OSS in one EU country and file quarterly, or block EU sales until ready.
UK VAT and others	UK, CA, AU…	UK digital services: from the first £ for non-UK sellers. Canada (GST/HST above CA$30k), Australia (GST above A$75k), and others have their own thresholds. Stripe Tax handles the calculation; registration and filing are still on us.
Income tax	Home country/state	Business income tax, Stripe sends 1099-K forms (US). Keep books from day one: revenue, node costs, refunds.
Sanctions (OFAC)	US	No service to sanctioned countries or people. Stripe screens payments; also block sign-ups from sanctioned regions.

11.6 Other rules worth knowing

📧 CAN-SPAM, PECR and ePrivacy

Service emails (resets, failed payments) are fine. **Marketing** emails need an unsubscribe link and our address (US). In the EU/UK they need opt-in consent. Keep the two lists separate.

🍪 Cookies

We only use necessary cookies (sign-in, account switcher), so no banner is required in the EU. If analytics or ad pixels are ever added, a consent banner is needed first.

💻 Computer Fraud and Abuse Act

DDoS, scanning and unauthorised access are crimes. Our Terms ban them. Keep logs to support police reports when we're attacked, and don't "hack back".

♿ Accessibility

The European Accessibility Act (from 28 June 2025) exempts micro-enterprises that provide services; US ADA claims against websites are common. The roadmap's accessibility pass lowers the risk cheaply.

™ Trademark

Search USPTO (and EUIPO) for "Tree Host" before spending on branding, and consider registering it once revenue allows.

📄 Provider terms

OVH, netcup and Hetzner terms make **us** responsible for our customers' actions. Our Terms must pass the same rules down (no mining, no attacks, no illegal content). ✓ They already do.

12 Legal readiness checklist

Checked against the current repository on 2 October 2026.

ITEM	STATUS	NOTES / NEXT STEP
Customer accepts Minecraft EULA (not us)	Done	Required before first start
"Not an official Minecraft product" disclaimer	Done	Site footer (`public/site.js`)
Acceptable use rules (no DDoS, mining, illegal content)	Done	Terms §3
Reasons and appeals for moderation actions (DSA-style)	Done	Action numbers, inbox, appeal once per action
Cancel online at any time	Done	Stripe Customer Portal. Also send a cancellation confirmation email
No tracking cookies	Done	Keep it that way, or add consent first
Governing law / legal entity name in Terms	To do	Set `TREEHOST_COMPANY` and `TREEHOST_JURISDICTION`
Age gate (13+ to sign up, 18+ or a parent to pay)	To do	Date-of-birth or checkbox at sign-up and checkout (COPPA)
Auto-renewal disclosure next to the pay button, and consent record	Partly	Add clear renewal text and a checkbox; store the time and IP of consent for 3 years
EU/UK 14-day withdrawal waiver at checkout	To do	Checkbox "Start my server now; I understand I lose my withdrawal right"
Refund policy with legal carve-outs	Partly	Add EU/UK/AU wording; clear SLA credits
DMCA agent registered and page published	To do	$6 at copyright.gov; add `/dmca` page and repeat-infringer clause
Lifetime plan wording	Review	Define "lifetime" and what happens if the service ends
Processors listed / DPAs signed	Partly	Privacy policy names the provider types; sign each provider's DPA, publish a subprocessor list
Customer DPA (players' data on their servers)	To do	Short DPA attached to the Terms
CSAM / law-enforcement process in staff docs	To do	Add to `staff-docs/support-playbook.md`
Tax registration and Stripe Tax	To do	Home state now; EU OSS / UK VAT before selling there
Terms and Privacy reviewed by a lawyer	To do	Roadmap item. Budget for a one-off review

13 Risk register

RISK	LIKELIHOOD	IMPACT	MITIGATION
Node overcommitted by multi-server plans	High	High	Count running servers against capacity before launch (§6.3)
DDoS takes all servers offline	High	High	OVH node with always-on filtering; panel behind Cloudflare; Minecraft proxy later
Single node hardware failure	Medium	High	RAID 1, nightly off-site backups, tested restore, documented rebuild (< 2 h with `install.sh`)
Further price increases at renewal	Medium	Medium	Unrepriced ranges, annual prepay when it's cheaper, pass increases on with 30 days' notice (Terms §7)
Lifetime plans become a permanent cost	Medium	Medium	Cap sales, raise the price, clarify the Terms
Stripe account review / reserve	Medium	High	Complete business profile, clear refund policy, low disputes, quick replies to Stripe
Under-13 users sign up	High	Medium	Age gate; delete under-13 data on discovery
Illegal or infringing content on a customer server	Medium	Medium	DMCA agent, notice-and-action, moderation tools, NCMEC process
Abuse reports ignored → provider blocks IP	Low	High	abuse@ alias monitored daily; reply within 24 h
Leaked secrets (Stripe key, Firebase key)	Low	High	`/etc` files at mode 640, never in git, Stripe restricted keys, rotate on staff changes

A Appendix & sources

Day-one command reference

Health check	`sudo bash /opt/tree-host/deploy/doctor.sh`
Update	`sudo bash /opt/tree-host/deploy/update.sh`
Live log	`journalctl -u tree-host -f`
Back up now	`sudo systemctl start tree-host-backup`
Settings	`/etc/tree-host.env` → `systemctl restart tree-host`
Data	`/var/lib/tree-host`

Starter monthly budget (Phase 1, RISE-S)

Node	$77 (+$77 setup, once)
Off-site backup	≈ $4–7
Domain renewal	≈ $1–3 (spread over a year)
Email, monitoring, Cloudflare	$0 (free tiers)
DMCA agent	$6 every 3 years
Total	≈ $85/mo, covered by about 19 Saplings

Decisions the team needs to make this month

#	DECISION	OPTIONS	RECOMMENDATION
1	Where to register the business	Founder's home state (simplest), Delaware/Wyoming LLC (popular, more filings), or outside the US	Home state LLC

#	DECISION	OPTIONS	RECOMMENDATION
2	First node	OVH RISE-S ($154 first month) · netcup RS 4000 G12 (≈$33) · Oracle Free (staging only)	RISE-S, or netcup if the fund is still $50
3	Fix for multi-server plans	Count running servers · limit to one running server · make sleep the default	Count running servers
4	Lifetime plan	Keep at $500 with a cap · raise to $750+ · pause sales until node 2	Cap at 2 per node and raise to $750
5	Who can buy	13+ with parent for payment · 18+ only · everyone	13+ accounts, 18+ (or parent) to pay
6	Selling to the EU/UK at launch	Register for EU OSS / UK VAT now · block EU/UK checkout until registered	US/Canada first, EU/UK when registered

Glossary

Bare metal / dedicated: a whole physical server rented to us alone. No neighbours, full CPU.

VPS / root server: a virtual slice of a bigger machine. "Dedicated cores" means no CPU sharing.

Node: a machine that runs customer Minecraft servers. Today Tree Host has one.

ECC RAM: memory that detects and fixes bit errors, protecting worlds from silent corruption.

DDoS: a flood of fake traffic meant to knock a server offline. Filtering ("scrubbing") removes it.

X3D: AMD CPUs with extra cache. Very good for Minecraft's main thread, but priced higher.

OOM killer: Linux stopping processes when memory runs out. It's what overselling RAM triggers.

MRR: monthly recurring revenue, the total of all active monthly plans.

DPA: Data Processing Agreement, the contract for handling personal data on someone's behalf.

Sources (accessed 2 October 2026)

Prices, stock and laws change: check every price on the order page before buying, and have a qualified lawyer and accountant confirm §11–12 before taking live payments. Tree Host is not affiliated with Mojang Studios or Microsoft.

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